

Siavash (Sæed) Kazemi

Resume

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Summary

- Data Scientist with 5+ years of experience in applied machine learning, data science, and research, supported by a broader background in engineering and software development.
- Experience developing production-oriented AI solutions using large language models (LLMs), machine learning, causal inference, and mathematical optimization.
- Strong experience in designing and evaluating AI-based decision systems, including data enrichment, entity profiling, classification, risk assessment, and model evaluation.
- Proficient in Python, PySpark, Pandas, Databricks, MLflow, Scikit-learn, TensorFlow, Snowflake, and MongoDB for data processing, experimentation, and machine learning workflows.
- Databricks Machine Learning Associate, Databricks Data Analyst Associate, and Microsoft Azure Data Scientist Associate certified.

Skills

- Technical Python, Pandas, PySpark, Databricks, Scikit-learn, TensorFlow, MLflow, Git, Snowflake, MongoDB, Azure, Databricks Lakehouse, Delta Lake, SQL
- Soft Skills Eager Learner, Teamwork, Highly Organized, Adaptability, Problem-solving, Time Management.

Related Work Experience

- 2024- to Present **Data Scientist**, *FIDDLEHEAD TECHNOLOGY*, Moncton, NB, Canada.
- Contribute to the design and development of AI and data-driven solutions for complex business and operational problems.
 - Apply machine learning, large language models, statistical analysis, and optimization techniques to improve automated decision-making systems.
 - Design scalable and explainable AI workflows by combining structured data, external information, business rules, and model-generated insights.
 - Develop data processing and analytical workflows using Python, Pandas, PySpark, Snowflake, and MongoDB.
 - Contribute to the development of optimization and causal inference solutions for data-driven planning and decision support.
- 2023- 2024 **Research Assistant**, *UNIVERSITY OF NEW BRUNSWICK*, Fredericton, NB, Canada.
- Contributed to data validation and quality assurance in a large-scale dataset.
- 2020-2023 **Research Assistant**, *UNIVERSITY OF NEW BRUNSWICK*, Fredericton, NB, Canada.
- Worked as part of a team on a federally funded, multi-partner project to develop the world's first pressure-based gait biometric system
 - Utilized various techniques such as pre-processing, feature extraction, and hyperparameters optimization to enhance the performance of the models
 - Implemented deep neural networks, including CNN, 1DCNN, and transfer learning techniques
 - Experience in working with Keras, Tensorflow, and scikit-learn libraries to build ML/DL models
- 2017-2019 **IT Support Specialist**, *MOEIN RAH GOSTAR KHORASAN COMPANY*, Mashhad, Iran.
- Train and supervise technical and non-technical staff
 - Provide guidance for purchasing of computer hardware, software, and supplies
- 2015-2017 **Electronic Engineer**, *SALMANIAN FARS CORPORATION*, Imam Khomeini highway, Isfahan, Iran.
- Designed and verified PCBs.
 - Worked with a variety of sensors and actuators.
 - Experienced in electronic design and integration.

2022-2023 **Teaching Assistant**, *UNIVERSITY OF NEW BRUNSWICK*, (1) Embedded System, and (2) Signals and Systems.

Selected Projects

- 2021 Implementing several approaches for time series classification as the project of "*Time Series Analysis*" course under the supervision of *Prof. Erik Scheme*.
- 2021 Implementing five ML algorithm as the project of "*Machine Learning and Data Mining*" course under supervision of *Prof. Huajie Zhang*.
- 2020 Comparing five algorithms for image registration as the project of "*Digital Image Processing*" course under the supervision of *Prof. Julian Meng*.
- 2014-2015 Doing research on Real-time target tracking algorithm based on machine vision for *Complex Research of Etebari*, located inside Isfahan University of Technology.

Educational Background

- 2023 **Master of Science in Electrical Engineering**, *University of New Brunswick, Canada*,
Courses: *Machine Learning & data Mining, Digital Image Processing, Intro to Pattern Recognition, Digital Signal Processing, Time Series Analysis, GPA – A+.*
Title of Thesis: *Exploring Performance Limits for Pressure-Based Gait Biometrics*
- 2014 **Master of Science in Communication**, *Isfahan University of Technology, Iran.*
Title of Thesis: *An Efficient Algorithm for Still and Moving Object Registration in Moving Video Camera Sequences*
- 2010 **Bachelor of Science in Electronics Engineering Technology**, *Shahid Rajaei Teacher Training University, Tehran, Iran.*
Title of Thesis: *Neural Network implementation by NEFPROX in order to approximate nonlinear function to use in medical applications*

Professional Certificate

Databricks Machine Learning Associate, certification by Databricks, [Credential Link](#).
Databricks Certified Data Analyst Associate, certification by Databricks, issued May 2026, [Credential Link](#).
DP-100: Azure Data Scientist Associate, certification by Microsoft, [Credential Link](#).
Watsonx Agentic AI Bootcamp – Level 2, certification by IBM, [Credential Link](#).
Databricks Accredited Azure Platform Architect, accreditation by Databricks, [Credential Link](#).
Machine Learning, online course by Standford University on coursera.org, Instructor: Professor Andrew Ng, [Credential Link](#).
Deep Learning, a 5-course specialization on coursera.org, Instructor: Professor Andrew Ng, [Credential Link](#).

Publications

- 2025 Larracy, R., Phinyomark, A., Salehi, A., MacDonald, E., **Kazemi, S.**, Bashar, S.S., Tabor, A. and Scheme, E., A DATASET OF HIGH-RESOLUTION PLANTAR PRESSURES FOR GAIT ANALYSIS ACROSS VARYING FOOTWEAR AND WALKING SPEEDS, *Scientific Data*. 2025 Aug 13;12(1):1415.
- 2023 **Kazemi, S.**, Phinyomark, A., Scheme, E., SAMPLE SIZE IN FLOOR SENSOR-BASED GAIT RECOGNITION FOR SMART HOME AND ACCESS CONTROL SCENARIOS, 2023 IEEE Sensors Applications Symposium, Ottawa, Canada, Jul 18-20, 2023.
- 2018 **S. Kazemi** and M. R. Ahmadzadeh, DPML-RISK: AN EFFICIENT ALGORITHM FOR IMAGE REGISTRATION, *International Journal of Engineering (IJE)*, In Press.

2016 **Saeed Kazemi** and M. R. Ahmadzadeh, PROVIDE EFFICIENT ALGORITHMS FOR MATCHING IMAGES BASE ON BRISK AND DIRECTIONAL PATTERN, journal of Iranian society of machine vision and image processing (in Farsi).

Volunteer Work

2021-2022 **Executive member**, *IRANIAN CANADIAN ASSOCIATION OF NEW BRUNSWICK*.

- Event planning and organizing for around 100 people
- Helping new international students and immigrants settle in New Brunswick